

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

Milestone 4 Documentation

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1. Introduction

The fourth milestone of the **VisionInspect AI – Manufacturing Defect Detection and Quality Inspection System** focuses on the final stage of the project, including system testing, model validation, performance optimization, deployment, documentation, and final demonstration.

During the previous milestones, the core VisionInspect AI platform was developed with functionalities for image upload, image preprocessing, AI-based defect detection, severity and risk assessment, inspection analytics, dashboard monitoring, and quality report generation.

In this milestone, the developed modules were tested as an integrated system to verify their functionality and usability. The platform was containerized using Docker to improve deployment consistency and portability. Cloud deployment was also configured using Render to make the application accessible through an online environment.

The main objective of this milestone was to validate the complete end-to-end workflow of the VisionInspect AI platform and prepare the system for final demonstration.

2. Objectives of Milestone 4

The major objectives completed during Milestone 4 are:

- Validate the defect detection and inspection workflow using different manufacturing images.
- Test the accuracy and consistency of AI-based inspection results.
- Evaluate the defect detection results, confidence scores, and severity/risk assessment workflow.
- Improve the usability and responsiveness of the dashboard and inspection result pages.
- Test the complete end-to-end workflow, from image upload and preprocessing to defect detection and report generation.
- Containerize and successfully test the VisionInspect AI frontend and backend using Docker in a local environment.
- Configure and deploy the application using Render.
- Demonstrate the complete VisionInspect AI platform through an end-to-end workflow.

3. System Testing and Validation

Testing and validation were performed to ensure that all modules of the VisionInspect AI platform work correctly as an integrated manufacturing quality inspection system.

The application was tested using manufacturing inspection images to verify the complete workflow from image upload to defect detection and report generation.

The testing process included:

- User authentication testing.
- Image upload testing.
- Image preprocessing validation.
- AI-based defect detection testing.
- Bounding box visualization testing.
- Defect and no-defect result validation.
- Severity and risk assessment testing.
- Dashboard data validation.
- Production quality report generation testing.
- PDF report download testing.

The objective of this testing process was to verify that the application provides consistent inspection results and that all major modules communicate correctly.

4. Defect Detection Validation

The AI-based defect detection module was tested using different manufacturing images from the dataset.

The VisionInspect AI system uses a YOLO-based computer vision model to identify defects within manufacturing images. When an image is uploaded, it is processed and passed to the trained model for prediction.

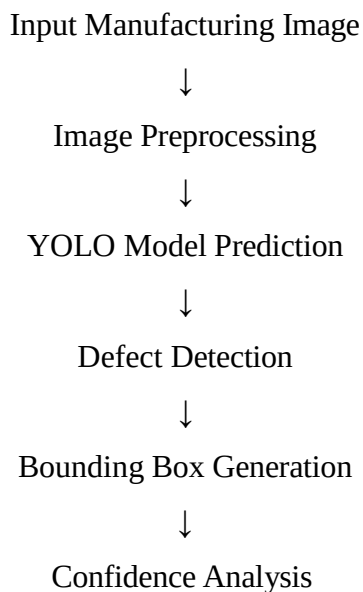
The model generates detection results containing information such as:

- Detected defect category.
- Detection confidence score.
- Defect location.
- Bounding box coordinates.

The detected defects are visually highlighted using bounding boxes, allowing the user to clearly identify the affected region of the manufacturing component.

Different input images were tested to evaluate whether the system could distinguish between defective and non-defective samples.

The validation workflow is shown below:





Defect / No Defect Result

This testing process helped validate the integration of the trained computer vision model with the backend inspection workflow.

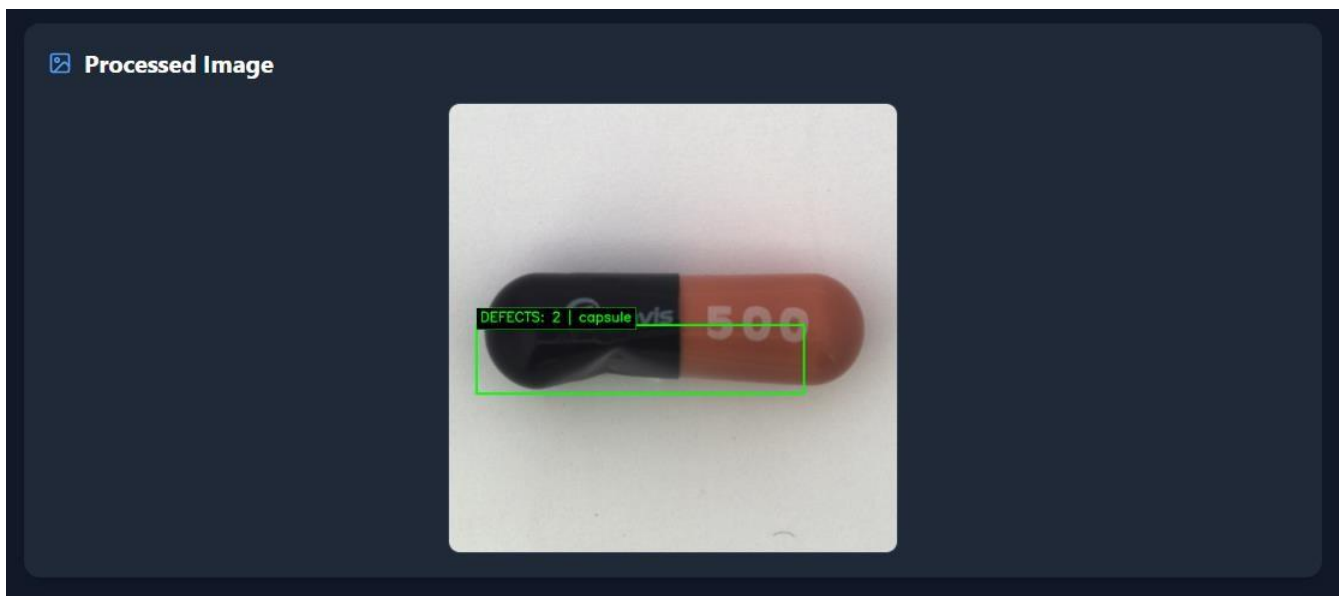


Figure 4.1: YOLO-Based Defect Detection and Bounding Box Validation

The figure demonstrates the AI-based detection result and visualization of the identified region using a bounding box.

5. Classification Quality and Severity Assessment

After detecting a defect, the system evaluates the confidence level of the prediction and generates a quality assessment.

The VisionInspect AI platform classifies the inspection result into appropriate risk and severity levels based on the detection confidence and inspection logic implemented in the backend.

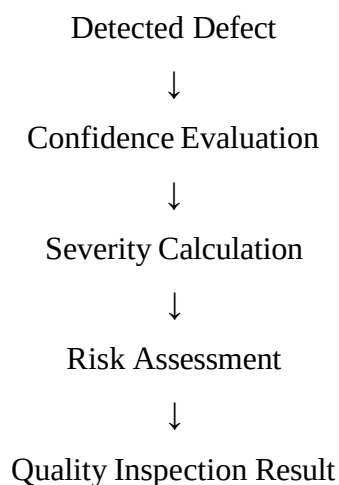
The severity assessment provides an additional level of quality intelligence beyond basic defect detection.

The system categorizes the inspection results into:

- High Risk
- Medium Risk
- Low Risk

This allows quality engineers to quickly understand the potential importance of a detected issue and prioritize inspection activities.

The classification and risk assessment workflow is represented as:

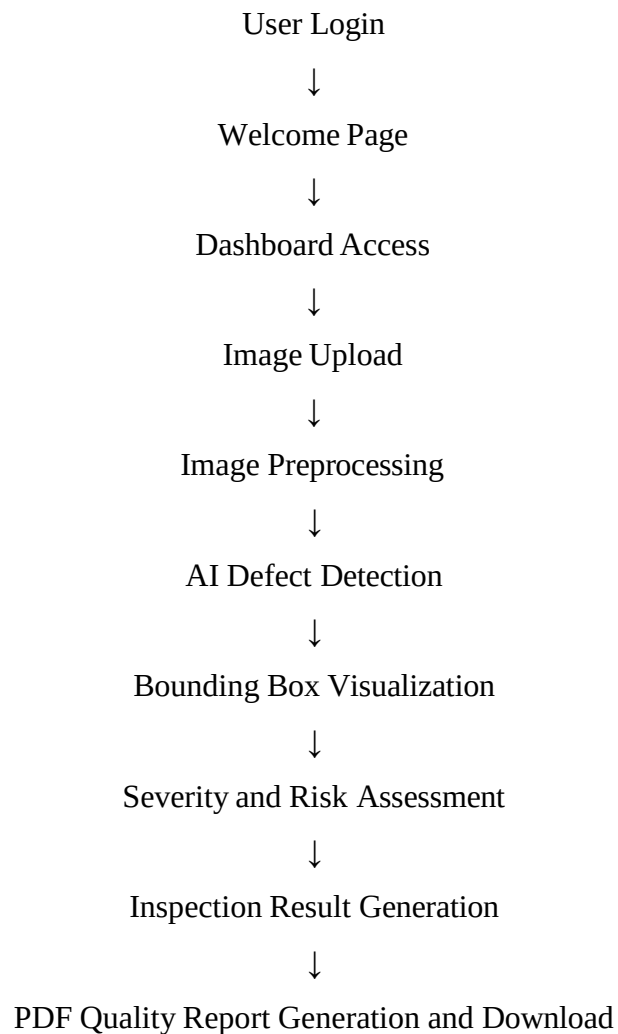


This feature improves the usability of the platform by converting AI prediction results into meaningful quality inspection insights.

6. End-to-End Functional Testing

The complete VisionInspect AI platform was tested as an integrated workflow.

The following end-to-end process was validated:



The testing confirmed that the major frontend, backend, and AI components work together to provide a complete manufacturing inspection workflow.

This end-to-end validation is important because the project integrates multiple technologies, including a React frontend, backend APIs, computer vision models, image processing modules, analytics dashboards, and report generation.

7. Model and Inspection Performance Optimization

Performance optimization was carried out to improve the overall efficiency of the defect detection workflow.

The following areas were considered during optimization:

- Optimization of the trained YOLO model for inference.
- Adjustment of confidence thresholds for defect detection.
- Efficient image preprocessing before model prediction.
- Testing and validation of the complete inspection workflow.

The confidence threshold was used to control which model predictions are considered during the inspection process.

This helped reduce unreliable detections and allowed the system to generate more meaningful inspection results.

The preprocessing pipeline also ensured that input images were prepared appropriately before being analyzed by the computer vision model.

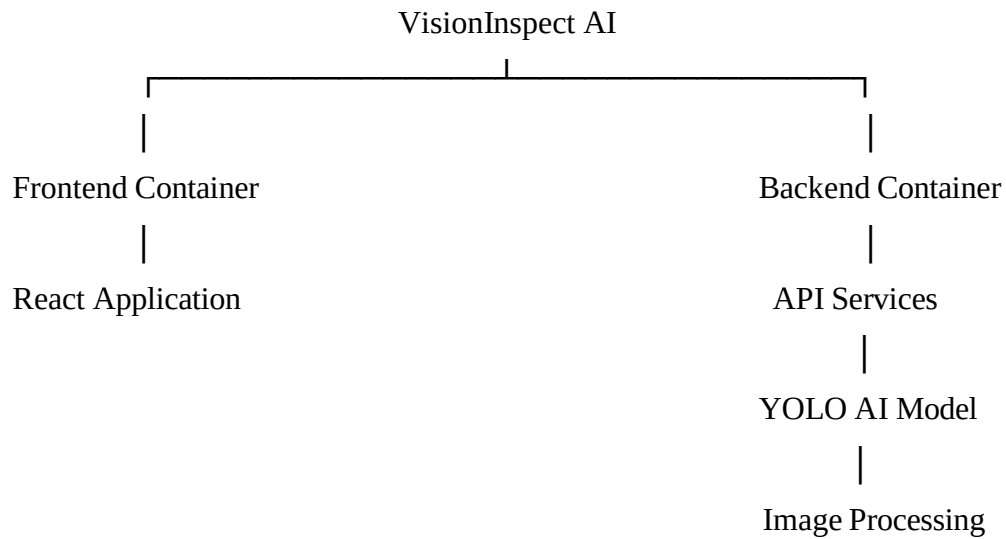
8. Docker Containerization

To improve deployment consistency and portability, the VisionInspect AI application was containerized using Docker.

Docker was used to package the application and its required dependencies into isolated containers.

The project includes separate environments for the frontend and backend components.

The containerized architecture can be represented as:



Docker provides several advantages for the project:

- Consistent application environment.
- Simplified deployment.
- Dependency isolation.
- Improved portability.
- Easier application execution across different systems.
- Better separation between frontend and backend services.

The Docker setup was successfully implemented and tested locally to verify that the application components could run in a containerized environment.



Figure 8.1: VisionInspect AI Docker Containers Running Successfully

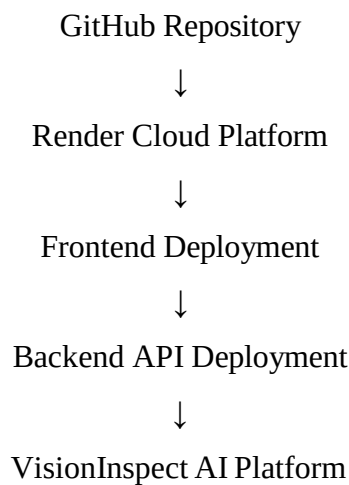
9. Cloud Deployment Using Render

After successful Docker containerization, the VisionInspect AI application was deployed using Render.

The purpose of cloud deployment is to make the VisionInspect AI platform accessible through an online environment rather than limiting execution to a local machine.

The deployment process includes configuring the application environment, connecting the project repository, setting required environment variables, and deploying the frontend and backend services.

The cloud deployment architecture can be represented as:



The deployed frontend and backend services were configured to enable communication between the web interface and the backend API in the cloud environment.

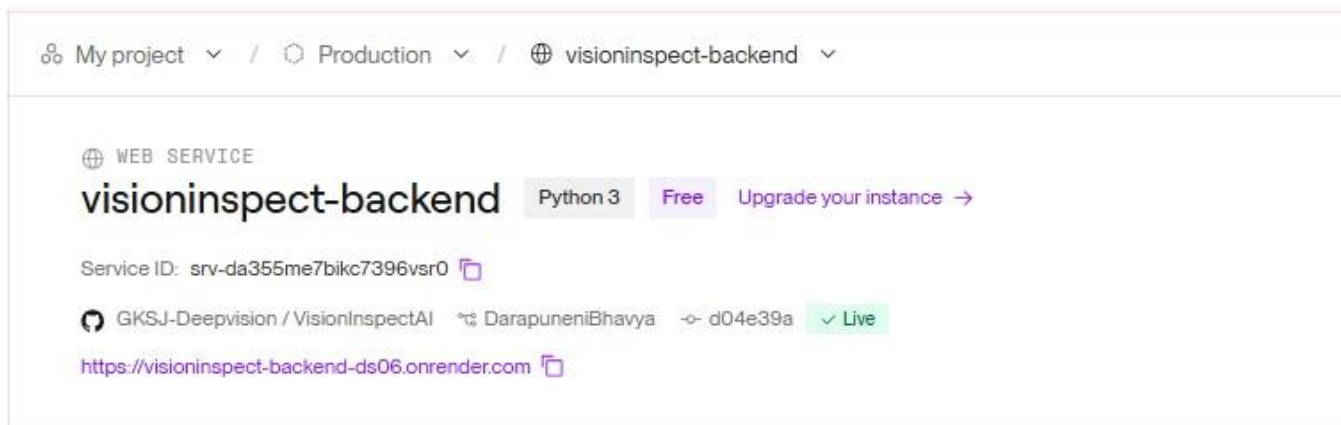


Figure 9.1: VisionInspect AI Backend Service Deployed on Render

The VisionInspect AI backend was deployed as a Python-based Web Service on Render. The deployed backend is connected to the project repository and is currently running in the production environment with a live service URL.



Figure 9.2: VisionInspect AI Frontend Deployed as a Render Static Site

The VisionInspect AI frontend was deployed as a Static Site on Render. The frontend is hosted independently and provides the web interface through a publicly accessible deployment URL.

This deployment enables the application to be accessed remotely and allows the frontend to communicate with the backend API through the deployed cloud environment.

10. Conclusion

Milestone 4 focused on the testing, validation, optimization, and deployment of the **VisionInspect AI – Manufacturing Defect Detection and Quality Inspection System**.

The complete application workflow was tested from user login and welcome page access to image upload, preprocessing, AI-based defect detection, bounding box visualization, severity and risk assessment, inspection result generation, and PDF quality report download. This testing confirmed the successful integration of the frontend, backend, and AI model components.

The defect detection workflow and classification quality were validated using manufacturing inspection images, while the inspection process was optimized to improve the overall performance and usability of the system. The application was also successfully containerized and tested in a local environment using Docker. In addition, cloud deployment using Render was configured to make the VisionInspect AI platform accessible through an online environment.

Overall, Milestone 4 helped validate VisionInspect AI as a complete end-to-end AI-powered manufacturing quality inspection platform, integrating computer vision, AI-based defect detection, quality assessment, reporting, and deployment.